

VMware Attack Simulation and Live Monitoring Setup

Objective

The objective of this task was to build a cybersecurity lab using VMware Workstation and Ubuntu Linux virtual machines to simulate real-world cyberattacks. The environment allowed attacks to be generated from one virtual machine and monitored in real time through the Digital Fort Live Monitor, providing practical exposure to attack detection and SOC monitoring.

How the Task Was Completed

To complete this task, VMware Workstation was installed along with the Ubuntu Linux ISO image. Two Ubuntu virtual machines were created and configured as VM1 and VM2. VM1 was connected to the Digital Fort Live Monitor using the provided API key, enabling it to forward security events to the monitoring dashboard. VM2 was used as the attacker machine to generate different attack activities against VM1. The generated events were successfully detected by the Digital Fort platform, allowing real-time observation and analysis of security alerts, attack patterns, and endpoint activity.

Challenges Faced

Creating the second virtual machine was initially difficult because of limited RAM and storage resources on the host computer. Multiple installation attempts failed due to insufficient system resources. After optimizing the available storage and receiving guidance from fellow interns, the second Ubuntu virtual machine was created successfully. Once both virtual machines were operational, the attack simulation environment functioned as expected.

Steps Performed

- Installed VMware Workstation.
- Downloaded the Ubuntu Linux ISO image
- Created two Ubuntu virtual machines (VM1 and VM2)
- Configured networking between the virtual machines
- Connected VM1 to the Digital Fort Live Monitor using the provided API key.
- Used VM2 to simulate attack activities against VM1.
- Observed and analyzed generated security events on the Live Monitor dashboard
- Verified that attack events were detected and logged successfully.

Skills Gained

- VMware Workstation setup and virtual machine management
- Ubuntu Linux administration
- Virtual networking concepts
- Cyberattack simulation in a controlled environment
- Security event monitoring and analysis

- Basic SOC workflow and live attack observation

Outcome

The VMware-based cybersecurity lab was successfully deployed and tested. Attack activities generated from the attacker virtual machine were detected by the Digital Fort Live Monitor, demonstrating how security monitoring solutions collect, process, and visualize events in real time. This task improved practical understanding of virtualized lab environments, attack simulation, and SOC monitoring processes.